Justin Ren Liu

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EDUCATION

University of California, Los Angeles (UCLA) **Expected June 2029**

Bachelor of Science, Electrical Engineering

- Relevant coursework: Signals and Systems, Computer Organization, Logic Design of Digital Systems.

SKILLS & TOOLS

- **Technical Skills:** Multi-layer PCB design, power systems, bare metal firmware, hardware integration, soldering.
- **Tools & Languages:** C/C++, Python, Verilog, Autodesk (CAD & PCB), KiCad, oscilloscope, Git, SolidWorks.

EXPERIENCE

Di Carlo Lab, California NanoSystems Institute at UCLA **August 2026 – Present**

Undergraduate Researcher

- Independently designed and tuned an ESP32 PI control loop regulating a 4-Peltier cold plate to within ±0.2°C of setpoint at steady state, integrated into eBot Bio's agentic AI platform for autonomous bioassays.
- Reworked the 3D-printed enclosure, lowering minimum cold plate temperature from 2 to -8°C.
- Built complete power harness off a switching supply, with low-side MOSFET switching for Peltier loads.

Mechatronics and Controls Lab, Mechanical and Aerospace Engineering at UCLA **June 2026 – Present**

Undergraduate Researcher

- Designed the main 2-layer PCB and pressure sensor board for a 6-DOF AUV, with cascaded buck-LDO power regulation with fuses, ESD protection, and P-MOSFET reverse-polarity protection, alongside a 9-axis IMU and ESP32–Raspberry Pi compute stack; validated power circuitry in LTspice.
- Worked with mechanical and systems subteams on a 10-person team to integrate electronics into the vehicle, and co-owned the electrical BOM across motors, ESCs, and batteries (sourcing, pricing, and cost reduction).
- Presented in weekly electronics meetings to non-electrical engineers on design updates and tradeoff choices.

UCLA Rocket Project — Valkyrie **November 2025 – June 2026**

Avionics Team Member

- Contributed to the design of multiple 4-layer ESP32-based avionics PCBs featuring LoRa telemetry, GPS, IMU, and dual altimeters for a hybrid rocket achieving ~13,400 ft apogee; assisted in hardware bring-up.
- Helped design and validate avionics power systems for a bodytube PCB (cascaded buck-LDO regulation, ideal-diode ORing), ensuring stable power delivery and seamless USB/battery power switching.
- Soldered and assembled avionics hardware, crimped and soldered board-to-board cable connections for avionics and EGSE systems, and performed debugging and board-level rework to resolve electrical issues.

PROJECTS

STM32 + FPGA Signal Processing Pipeline **October 2026**

- Implemented STM32F446 peripherals (GPIO, UART, timers, interrupts) via direct register access; achieved 1 kHz ADC sampling using timer-triggered DMA into a 1000-sample buffer with minimal CPU overhead.
- Streamed samples over UART to a Tang Nano 20K, which reassembles them with a Verilog frame parser, applies an 8-sample moving-average filter, and forwards raw and filtered data to a PC for live plotting.
- Wrote and simulation-verified UART RX/TX modules (baud-rate generator, FSMs, RX input synchronizer).

Vitagotchi — UCLA IdeaHacks 2026 (36h, team of 4) **April 2026**

- Developed a 2x ESP32 wearable system for real-time health state classification (accelerometer, heart rate, temperature/humidity), with ESP-NOW wireless communication, Firebase data logging and web dashboard.
- Designed the desktop companion unit (interactive plush form factor), integrating dual OLED displays, RGB LED, flex sensors, and LLM-generated responses to provide real-time, multimodal user interaction.
- Built portable power systems (boost and LDO regulation) and rapidly prototyped hardware on perfboard.

Custom ESP32 Development Board **February 2026**

- Designed and routed a 2-layer ESP32 development board from component selection to board bring up.
- Routed USB D+/D– as a matched differential pair, placed decoupling capacitors, ground stitching vias, implemented external BOOT and RESET switches and added reverse polarity and ESD protection.